

SIMATS ENGINEERING
CO- 2: AT – 2: SIMULATION TASK - MEDIUM
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COURSE NAME: 5G / 6G COMMUNICATION SYSTEMS
COURSE CODE: ECA5608

17. Simulation of 5G NR air interface performance under different numerologies, bandwidths, latency, subcarrier spacing, and throughput (BL4)

AIM:

To analyze the performance of the 5G NR air interface under different numerologies, bandwidths, subcarrier spacing, latency, and throughput using MATLAB.

SOFTWARE USED:

MATLAB

MATLAB CODE:

```
clear;
```

```
clc;
```

```
Bandwidth = [20 50 100];    % MHz
```

```
SCS = [15 30 60];          % kHz
```

```
Numerology = [0 1 2];
```

```
SlotDuration = 1./(2.^Numerology);
```

```
disp(SlotDuration)
```

```
Efficiency = 6;

Throughput = Bandwidth*Efficiency;

disp(Throughput)

figure;

plot(Bandwidth,Throughput,'-o','LineWidth',2);

xlabel('Bandwidth (MHz)');

ylabel('Throughput (Mbps)');

title('Bandwidth vs Throughput');

grid on;

Latency = SlotDuration;

figure;

plot(Numerology,Latency,'-s','LineWidth',2);

xlabel('Numerology');

ylabel('Latency (ms)');

title('Numerology vs Latency');

grid on;

figure;

plot(SCS,SlotDuration,'-*','LineWidth',2);

xlabel('Subcarrier Spacing (kHz)');

ylabel('Slot Duration (ms)');
```

```
title('SCS vs Slot Duration');
```

```
grid on;
```

```
LatencyBW = [1 0.8 0.5];
```

```
figure;
```

```
bar(Bandwidth,LatencyBW)
```

```
xlabel('Bandwidth')
```

```
ylabel('Latency')
```

```
title('Bandwidth vs Latency')
```

OUTPUT:





